

Solve equations with brackets

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Solve $6(x + 4) = 72$. Copy or complete the first step.

2. A number is tripled and 14 is added to make 65. Find the number.

3. Do $4x + 12 = 40$ and $2x + 6 = 20$ have the same solution?

4. Miro says: Miro changes one side of an equation without making the same balanced change to the other side. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Solve $6(x + 4) = 72$. Copy or complete the first step.

Answer $x = 8$.

2. A number is tripled and 14 is added to make 65. Find the number.

Answer 17.

3. Do $4x + 12 = 40$ and $2x + 6 = 20$ have the same solution?

Answer Yes, both give $x = 7$.

4. Miro says: Miro changes one side of an equation without making the same balanced change to the other side. Is this correct? Explain.

Answer Use an inverse operation on the whole equation and check by substitution.

Solve equations with brackets

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Solve $6(x + 4) = 72$.

6. Check this result using an inverse, estimate or known fact: Do $4x + 12 = 40$ and $2x + 6 = 20$ have the same solution?

2. A number is tripled and 14 is added to make 65. Find the number.

7. Prove or explain your reasoning: Solve $x + 38 = 91$.

3. Do $4x + 12 = 40$ and $2x + 6 = 20$ have the same solution?

8. Identify and correct this misconception: Miro changes one side of an equation without making the same balanced change to the other side.

4. Solve this independently and show every stage: Solve $6(x + 4) = 72$.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: A number is tripled and 14 is added to make 65. Find the number.

10. Write and solve a similar question to: Do $4x + 12 = 40$ and $2x + 6 = 20$ have the same solution?

Teacher answers and checking prompts

1. Solve $6(x + 4) = 72$.
Answer $x = 8$.
2. A number is tripled and 14 is added to make 65. Find the number.
Answer 17.
3. Do $4x + 12 = 40$ and $2x + 6 = 20$ have the same solution?
Answer Yes, both give $x = 7$.
4. Solve this independently and show every stage: Solve $6(x + 4) = 72$.
Answer $x = 8$.
5. Use a second method or representation for: A number is tripled and 14 is added to make 65. Find the number.
Answer Expected result: 17. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Do $4x + 12 = 40$ and $2x + 6 = 20$ have the same solution?
Answer Expected result: Yes, both give $x = 7$. A valid check must be shown.
7. Prove or explain your reasoning: Solve $x + 38 = 91$.
Answer $x = 53$. The explanation must show why the method works.
8. Identify and correct this misconception: Miro changes one side of an equation without making the same balanced change to the other side.
Answer Use an inverse operation on the whole equation and check by substitution.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Do $4x + 12 = 40$ and $2x + 6 = 20$ have the same solution?
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Solve equations with brackets

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Solve $x + 38 = 91$.

6. Create a new example that tests the same idea as: Do $4x + 12 = 40$ and $2x + 6 = 20$ have the same solution?

2. Prove the answer to this question, rather than only calculating it: Solve $6(x + 4) = 72$.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: A number is tripled and 14 is added to make 65. Find the number.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: Yes, both give $x = 7$.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro changes one side of an equation without making the same balanced change to the other side.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Solve $x + 38 = 91$.
Answer $x = 53$. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Solve $6(x + 4) = 72$.
Answer Expected result: $x = 8$. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: A number is tripled and 14 is added to make 65. Find the number.
Answer Expected result: 17. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: Yes, both give $x = 7$.
Answer One valid reconstruction is: Do $4x + 12 = 40$ and $2x + 6 = 20$ have the same solution? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro changes one side of an equation without making the same balanced change to the other side.
Answer Use an inverse operation on the whole equation and check by substitution.
6. Create a new example that tests the same idea as: Do $4x + 12 = 40$ and $2x + 6 = 20$ have the same solution?
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.